



Tecnológico de Monterrey

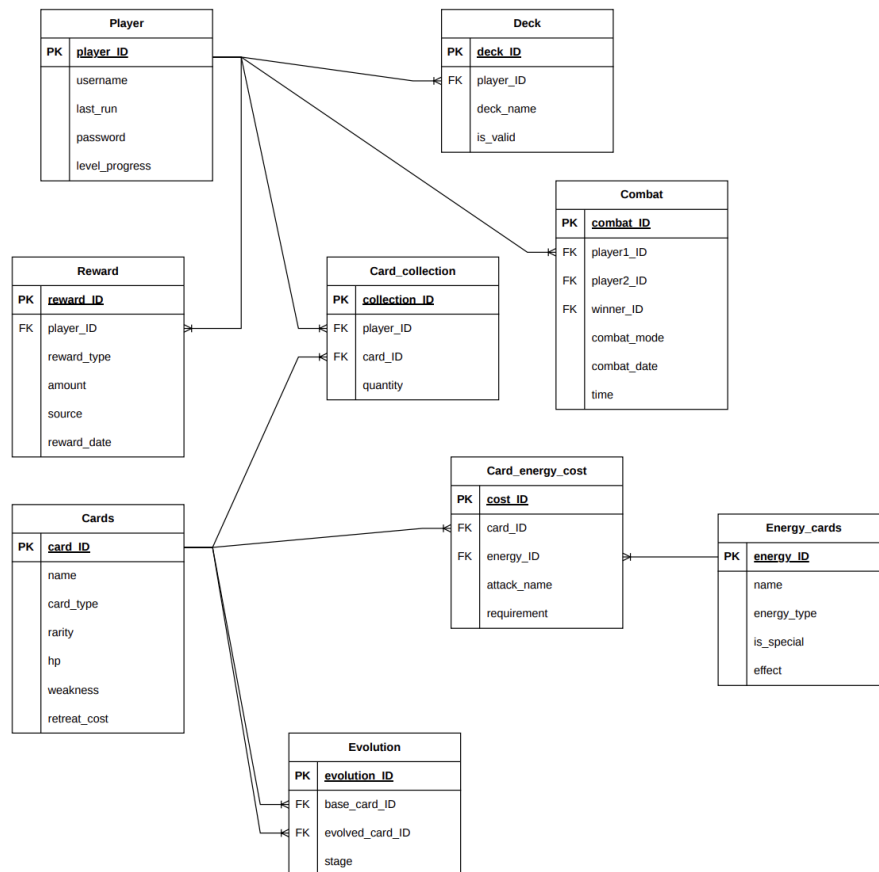
Pokemon TCG Database Modeling Exercise

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Software Construction and Decision Making

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Player

This table acts as the primary registry for the players within the system, tracks the player's progression as well as the last time they accessed the game. This date is essential for personalization of the player's experience and security protocols.

Cards

The cards table acts as the catalog for every playable card in the game, excluding energies. It stores the attributes of every card such as card's name, its specific type, rarity level, health points, elemental weaknesses and the cost required to use it during a combat.

Energy_cards

This table is dedicated exclusively to the energy resources, categorizing the different types of energy available in the environment of the game. By separating the energies from the main card table the system can manage special effects more efficiently and link those resources to the requirements of every pokemon attack without duplicating the data.

Card_energy_cost

This is the technical table that links the pokemon cards to their necessary energy requirements, it specifies the name of every attack and the exact type and quantity of energy is required for its use. Allowing for complex cards that might require multiple and different energy types to activate..

Deck

Manages the logical groups of cards the players assemble every play. Each record represents a specific deck owned by a player including the name and validation ensuring that only decks meeting the game's rules can be used in combat.

Card_collection

Records the specific collection for each player, identifying which unique cards they own and how many copies are available in their inventory, helping with the deck building capabilities of the game.

Combat

The combat table acts as a historical log for every combat played in the game. It captures critical data including the IDs of both players, the ID of the winner, game mode selected (casual or competitive) and the precise time and date of the event. This information is essential for players ranking and combat history.

Reward

Tracks all the assets, currency or items granted to the players through different game's activities, connects each reward to a specific player and lists what they got, how much they receive and the reason for it. Keeping a clean and safe history of items and currencies every player collects.

Evolution

This table defines the connection of different pokemon stages in the game. It establishes which card is the base form and which is the evolved result. This logic is essential to ensure the gameplay rules that prevent evolved pokemon from being played without their previous stages.